PIC09918A-V2.0_PRO

https://github.com/visrealm/pico9918/wiki/Schematics#socket-map

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* Wired directly, not through this GAL:
*   bus A0     -> VDP pin 13 (MODE)
*   bus A1     -> VDP pin 11 (MODE1)  (or GND)
*   bus /RESET  -> VDP pin 34
*   VDP /INT    -> SBC IRQ6 (socket pin 16)
*   6502 D0..D7  -> VDP pins 17..24 (CD7..CD0, reversed: D7 -> pin 24)
*   '573: D <- AVR bus, Q -> 6502 bus, LE <- AVR PD5, /OE <- nKBRD
*   TX latch: see below.
*
* Rev 06 change - the serial TX latch is clocked straight from PHI2:
*   nDISPWE (pin 21) is now "write cycle to $7F1E" with NO PHI2 term. It is
*   settled during PHI2 low, long before PHI2 falls at the end of the cycle.
*   The latch clock edge is then made from PHI2 by a fast gate on the card,
*   a few ns after PHI2 falls - inside the 65C02's ~10 ns write data hold -
*   so GAL speed no longer matters for this latch. Two ways to build it:
*
*   A) 74HCT574 (keep it), CLK <- 74ACT02 NOR(PHI2, nDISPWE), /OE <- /DOE
*      CLK rises when PHI2 falls during a $7F1E write.
*      It also rises once early in that write cycle (when nDISPWE goes
*      low while PHI2 is still low), latching junk that the real edge
*      then overwrites - harmless, as the handshake guarantees the AVR
*      has already taken the previous byte.
*      One gate, no other new chips.
*
*   B) 74HCT377, /E <- nDISPWE, CLK <- PHI2 via 74ACT04/74F04 inverter,
*     plus a 74HCT541 between the '377 Q outputs and the AVR bus
*     (/OE <- /DOE), because the '377 has no output enable.
*      Cleanest: exactly one clock edge per cycle, no early edge.
*      Costs two extra chips.
*/

/* ------------------- Inputs -------------------- */
PIN 1  = PHI2;     /* bus CLK_IN (PHI2), also the register clock */
PIN 2  = nIOR;     /* /IOr from main GAL - not used in rev 05  */
PIN 3  = nIOW;     /* /IOw from main GAL (address only)       */
PIN 4  = nIO_EN;   /* /IO_EN, low when A15-A8 = $7F          */
PIN 5  = A7;
PIN 6  = A6;
PIN 7  = A5;
PIN 8  = A4;
PIN 9  = A3;
PIN 10 = A2;
PIN 11 = A1;
PIN 13 = A0;
PIN 14 = KBAVAIL;  /* from AVR PC5, active high (I/O pin as input) */
PIN 23 = nDOE;     /* from AVR PD4, low while AVR reads '574     */

/* ------------------ Outputs ------------------- */
PIN 15 = nKBIRQ;   /* to SBC IRQ5, active low             */
PIN 16 = D7;       /* to 6502 data bus D7, tri-state       */
PIN 17 = nKBRD;    /* to '573 /OE and AVR PD2 (INT0)        */
PIN 18 = nCSW;     /* to VDP pin 14                       */
PIN 19 = nCSR;     /* to VDP pin 15                       */
PIN 20 = D6;       /* to 6502 data bus D6, tri-state       */
PIN 21 = nDISPWE;  /* TX latch enable (no PHI2) - see rev 06   */
PIN 22 = DBUSY;    /* registered, to AVR PD3              */
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